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ORIGINAL ARTICLE

Anxiety and depression among reproductive-aged women in Bangladesh: burden, determinants, and care-seeking practices based on a nationally representative demographic and health survey

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Abstract

Purpose This study aimed to investigate the prevalence, determinants, and care-seeking practices for anxiety and depression among reproductive-aged women in Bangladesh.

Methods We utilized data from the 2022 Bangladesh Demographic and Health Survey, with a total weighted sample of 20,029 women aged 15–49. Anxiety and depression symptoms were assessed using the Generalized Anxiety Disorder-7 (GAD-7) scale and the Patient Health Questionnaire (PHQ-9) scale, respectively. Prevalence was reported with 95% confidence intervals (CI). Multiple logistic regression was conducted to identify associated factors, and the results were presented as adjusted odds ratios (AOR) with 95% CI.

Results Among women, 4% (95% CI: 3.9%, 4.9%) had symptoms of moderate to severe anxiety, and 5% had moderate to severe depression. The prevalence of both symptoms was significantly higher ($p < 0.05$) in Khulna, Rangpur, and Sylhet. Older women (45–49 years) (anxiety: AOR: 2.33, 95% CI: 1.55, 3.50; depression: AOR: 1.63, 95% CI: 1.15, 2.31) and women who made all major household decisions (anxiety: AOR: 1.30, 95% CI: 1.00, 1.69; depression: AOR: 1.41, 95% CI: 1.07, 1.80) were more likely to experience moderate to severe anxiety and depression. Among women with symptoms of moderate to severe anxiety or depression, 22% (95% CI: 19.9%, 25.0%) sought care, and 8% (95% CI: 6.6%, 10.1%) used medication.

Conclusion The findings highlight a significant burden of anxiety and depression among women in Bangladesh, with notable regional disparities. These insights underscore the need for targeted interventions to address regional disparities, support vulnerable groups such as older women and those involved in household decision-making, and integrate mental health services into primary healthcare for better mental well-being among women in Bangladesh.

Article highlights

- In Bangladesh, 4% and 5% of reproductive-aged women experienced moderate to severe symptoms of anxiety and depression respectively.
- Highest prevalence of anxiety and depression was in Khulna, Rangpur, and Sylhet.
- Women with older age and higher decision-making autonomy are associated with anxiety and depression.
- 22% of women with symptoms of anxiety and depression sought care from a healthcare provider and 8% took medication.

Keywords Anxiety · Depression · Mental health · Reproductive age · Women · Bangladesh

Introduction

Mental health is a state of well-being that enables individuals to manage life's stressors, realize their potential, learn and perform effectively, and contribute to their communities (World Health Organization 2022a). Deviations from this state can lead to mental disorders, which disrupt thought processes, emotional regulation, and behavior, necessitating

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targeted support and treatment (World Health Organization 2022b). Mental disorders consistently rank among the top ten global health burden, affecting approximately one in eight individuals worldwide (GBD 2019 Mental Disorders Collaborators 2022; World Health Organization 2022b). In 2021, an estimated 970 million people, including 236 million in South Asia, were affected by mental disorders (GBD 2019 Mental Disorders Collaborators 2022). These disorders significantly contribute to disability and economic productivity loss globally (Whiteford et al. 2013). The age-standardized disability-adjusted life-years (DALYs) rate for mental disorders was 1703.3 per 100,000 females and 1426.5 per 100,000 males worldwide (GBD 2019 Mental Disorders Collaborators 2022). The economic burden of lost productivity due to depression and anxiety is estimated at US$1 trillion per year, with projections reaching $16 trillion by 2030 (The Lancet Global Health 2020).

Anxiety and depression are among the major global mental health concerns. Generalized anxiety disorders are characterized by excessive worry, fear, and related behavioral abnormalities, often severe enough to cause substantial distress or functional impairment (World Health Organization 2022b). Depression is marked by long-term low mood, loss of enjoyment, and lack of interest in activities (World Health Organization 2019b, 2022b). Globally, 7% of the population has anxiety disorders, while 4% experience depressive disorders (GBD 2021 Results 2022). Projections indicate that by 2050, depressive disorders will be the 7th leading cause of disease burden, while anxiety disorders will rank 19th (GBD 2021 Forecasting Collaborators 2024). Both conditions cause severe morbidity, disability, reduced productivity, and economic burden.

The Lancet Commission on Global Mental Health and Sustainable Development emphasizes that mental health is a fundamental human right and essential for the development of all countries, particularly in low- and middle-income countries (LMICs) (Patel et al. 2018). In Bangladesh, a lower-middle-income country in South Asia, 5% of the population is affected by anxiety disorders, and 6% experience depressive disorders (GBD 2021 Results 2022). However, these estimates are model-based. A cross-sectional study in Bangladesh found that 27% of female university students experienced anxiety and 36% experienced depression during the COVID-19 pandemic (Zabun Nahar et al. 2022). Furthermore, the prevalence of anxiety and depression symptoms is higher among women compared to men (18.8% vs. 12.5%) (Hasan et al. 2021). Despite evidence of a high burden of mental health issues in Bangladesh, most studies are geographically and condition-specific, leaving a gap in nationally and regionally representative estimates.

Experts advise that women experiencing moderate to severe anxiety and depression should seek professional assistance (Afifi 2007). However, multiple factors, including socioeconomic constraints, low educational attainment, reduced decision-making autonomy, inadequate awareness of mental health conditions, and concerns about stigma and potential repercussions for marital stability, often impede their utilization of mental health services (Goyal et al. 2020; Dutta et al. 2022). In Bangladesh, women frequently exhibit insufficient knowledge about mental health disorders, their prevention, management, and available referral systems, and they continue to face these barriers (Jorm 2000; World Health Organization 2019a). The scarcity of mental health professionals, the uneven distribution of facilities to urban areas, and the lack of effective stewardship in implementing adequate mental health policies exacerbate this situation (Islam 2015).

For the first time, the Bangladesh Demographic and Health Survey (BDHS) has included a mental health module. This paper aims to provide a nationally representative estimate of the prevalence of anxiety and depression among reproductive-aged women, alongside an analysis of the underlying factors contributing to this burden and the associated care-seeking practices.

Methods

Data source and sampling technique

This study utilized data from the BDHS, a nationally representative survey conducted between August and December 2022. The survey is implemented under the Demographic and Health Survey (DHS) program established by the United States Agency for International Development (USAID), which has been conducted across 90 LMICs (USAID 2023). The BDHS 2022 data is publicly accessible upon request at https://dhsprogram.com/methodology/survey/survey-display-584.cfm.

The BDHS 2022 employed the Integrated Multi-Purpose Sampling Master Sample, based on the 2011 population census by the Bangladesh Bureau of Statistics. A two-stage stratified sampling procedure was used: 675 enumeration areas (EAs) were selected with probability proportional to size, and an average of 45 households per EA were systematically chosen. Eligible women in each household were surveyed about their background and reproductive history. Among 45 households, 30 were selected for interviewing with the full questionnaire, while 15 were selected for the shortened version (NIPORT and ICF 2024).

Initially, 30,330 households were selected for the BDHS 2022 sample. Of these, 30,149 were found to be occupied, and 30,018 were successfully interviewed, resulting in a response rate of 99.6%. Within these households, 30,358 ever-married women aged 15–49 were identified as eligible for individual interviews. Of these, 30,078

participated, resulting in a 99.1% response rate. For the detailed individual questionnaire that includes the mental health module,20,217 women were eligible, and 19,987 were successfully interviewed, with a response rates of 98.9%. Finally, 20,029 (weighted) women aged 15–49 were included in our analysis (Figure S1 in supplementary material A).

Outcome variables

The primary outcome variables were symptoms of anxiety and depression. Anxiety was measured using the Generalized Anxiety Disorder Scale (GAD-7), which is a 7-item self-reported tool to assess the occurrence and intensity of generalized anxiety symptoms over the previous two weeks. Women's responses were recorded on a four-point Likert scale from 0 (never) to 3 (always), resulting in total scores ranging from 0 to 21 (Spitzer et al. 2006). GAD-7 score is categorized as follows: 0–4 indicate none, 5–9 mild, 10–14 moderate, and 15–21 severe anxiety symptoms. A score of 10 or above was categorized as having symptoms of moderate to severe anxiety (Spitzer et al. 2006). The GAD-7 scale demonstrated strong reliability with a Cronbach's α of 0.834. Depression symptoms were assessed using the Patient Health Questionnaire (PHQ-9), a self-reported instrument comprising 9 items evaluating symptoms of depression over the last two weeks (Kroenke et al. 2001). Similar to anxiety, women's responses were recorded on a four-point Likert scale, resulting in total scores ranging from 0 to 27, delineating the following severity levels: no depression (scores 0–4), mild (scores 5–9), moderate (scores 10–14), moderately severe (scores 15–19), and severe (scores 20–27). A score of 10 or above was categorized as symptoms of having moderate to severe depression (Kroenke et al. 2001; Banik et al. 2022). The PHQ-9 scale also showed strong internal consistency with a Cronbach's α of 0.821. Previous studies in Bangladesh also used the same cut-off score of 10 on both the PHQ-9 and GAD-7 to assess the presence of depression and anxiety symptoms, respectively (Banik et al. 2022; Wahid et al. 2023). The details of these scales are given in Section 1 of the supplementary material A.

Respondents with scores above 10 on the PHQ-9 or GAD-7 scales were asked if they had ever sought help for mental health issues. Respondents who sought help specifically from a healthcare provider were considered as sought care while those who sought help from informal sources or did not seek help at all were categorized as didn't seek care (Shawon et al. 2024; Pandey et al. 2024). Additionally, we assessed medication use by considering respondents who reported taking prescribed medication for anxiety or depression within the past two weeks.

Explanatory variables

We utilized several explanatory variables covering socio-economic and demographic factors, maternal and paternal characteristics, household attributes, child-related factors, and regional characteristics. Socio-economic and demographic factors included women's age (15–19, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49), education level (no education, primary, secondary, higher secondary or above), occupation (housewife, agriculture, non-agriculture, officer/businessmen, retired/others), husband's age (15–19, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49, 50+), husband's occupation (not working, agriculture, non-agriculture, officer/businessmen, retired/others), marital status (currently married), and religion (Islam, others). Household attributes included the wealth index (poorest, poorer, middle, richer, richest), household head's sex (male, female), current co-residence with husband [living with her, living elsewhere (within Bangladesh or not in Bangladesh)], number of household members (less or equal 4, 4+), mass media exposure (no, yes), mobile phone ownership (no, yes). The wealth index, constructed by the Demographic and Health Survey (DHS) program, is a composite measure of a household's cumulative living standard, calculated using principal component analysis (PCA) (Rutstein 2015). Mass media exposure was defined as a composite variable and respondents were considered to have media exposure if their household owned either a radio or a television (Ferdausi et al. 2022; Akter et al. 2023). Furthermore, we assessed women's household decision-making autonomy across variables from four key areas: respondent's healthcare, large household purchases, visits to family or relatives, and the use of the husband's earnings (Acharya et al. 2010). Women were assigned a score of '1' if they made decisions either alone or jointly with their husbands in each domain and a '0' if the decision was made solely by the husband or someone else (total score ranged from 0 to 4). Additionally, child-related factors included the number of children (0, 1, 2, 3, 4+) and under-5 children (0, 1, 2+). Additionally, we considered current pregnancy status (no, yes) and current breastfeeding status (no, yes). Regional-level variables included division (Barishal, Chattogram, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet) and residence area (urban or rural).

Statistical analysis

Data cleaning, recoding, and analysis were conducted using Stata version 17.0 (StataCorp LP, College Station, Texas) in accordance with DHS guidelines (Croft et al. 2020). To ensure the survey's representativeness, sample weights were applied, and the complex survey design was adjusted by accounting for the primary sampling unit (PSU) and

strata. The Stata "Svyset" command was utilized to handle the complex survey design. Additionally, we adhered to the STROBE cross-sectional reporting guidelines (von Elm et al. 2014).

Descriptive statistics were used to summarize the characteristics and report the outcomes with a 95% confidence interval (CI), including frequencies and row percentages for categorical variables. The chi-squared test was performed to identify relationships between study variables and the outcome variable. Variables that showed a significant association in bivariate analysis were included to the multivariable logistic regression model to identify factors associated with mental health outcomes among women aged 15 to 49 years. The adjusted odds ratio (AOR) with a 95% CI was used to determine the strength of the associations.

Result

Among the 20,029 women, 18% ($n=3,523$) were aged 25–29. Almost 60% ($n=11,062$) of them had secondary education or higher, and 90% ($n=18,107$) were Muslims. More than two-thirds of them ($n=13,940$) of them owned a mobile phone. Around 95% ($n=19,060$) of women were currently married, and the majority (63%) were housewives. Approximately 6% ($n=1,181$) of them were pregnant, 48% ($n=9,597$) had at least one child under five years of age, and 20% ($n=4,073$) were breastfeeding during the time of the survey. Around half of them ($n=10,017$) made all four major household decisions (Table 1).

Regarding each symptom of anxiety, 11% of women ($n=2,241$) reported that they felt nervous, anxious, or on edge often or all the time. Around 8% of women ($n=1,665$) felt they were worrying too much about different things often or all the time. Around 7% of women ($n=1,402$) felt that they were easily becoming annoyed or irritated often or all the time. Regarding each symptom of depression, 10% of women ($n=2,003$) reported feeling tired or having little energy often or all the time. Around 9% of women ($n=1,803$) reported trouble falling or staying asleep or sleeping too much often or all the time. Around 7% of women ($n=1,402$) experienced feelings of being down, depressed, or hopeless often or all the time (Fig. 1). Associated factors of these selected items of GAD-7 and PHQ-9 is provided in Table S1 of the supplementary material B.

Approximately 4% (95% CI: 3.9%, 4.9%) of women had symptoms of moderate to severe anxiety, and 5% (95% CI: 4.4%, 5.4%) had symptoms of moderate to severe depression. Approximately 7% of them (95% CI: 6.3%, 7.5%) had either anxiety or depression, while 2.4% (95% CI: 2.1%, 2.8%) experienced both concurrently. For anxiety, an additional 22% of them had symptoms of mild anxiety. For depression,

Table 1 Distribution of study variables among reproductive-aged women in Bangladesh ($n=20,029$)

Variables	*Weighted frequency* (%)
Women Age	
15–19	1729 (8.6)
20–24	3289 (16.4)
25–29	3523 (17.6)
30–34	3437 (17.2)
35–39	3344 (16.7)
40–44	2546 (12.7)
45–49	2160 (10.8)
Women Education	
No education	2752 (13.7)
Primary	5214 (26)
Secondary	9359 (46.7)
Higher secondary or above	2703 (13.5)
Husbands' Age	
15–19	115 (0.6)
20–24	1038 (5.2)
25–29	2331 (11.6)
30–34	2830 (14.1)
35–39	3356 (16.8)
40–44	3004 (15)
45–49	2398 (12)
50 +	3986 (19.9)
Missing	969 (4.8)
Husbands' Education	
No education	4083 (20.4)
Primary	5386 (26.9)
Secondary	6196 (30.9)
Higher secondary or above	3356 (16.8)
Missing	1006 (5)
Women's Occupation	
Housewife	12,562 (62.7)
Agricultural	4723 (23.6)
Non-agricultural	1930 (9.6)
Officer/Businessmen	645 (3.2)
Retired/Others	168 (0.8)
Husbands' Occupation	
Not working	660 (3.3)
Agricultural	4632 (23.1)
Non-agricultural	8720 (43.5)
Officer/Businessmen	4948 (24.7)
Retired/Others	99 (0.5)
Missing	969 (4.8)
Marital status	
Currently married	19,060 (95.2)
Separated/Deserted/Divorced/Widowed	969 (4.8)
Religion	
Islam	18,107 (90.4)
Others	1921 (9.6)

Table 1 (continued)

Variables	*Weighted frequency* (%)
Wealth Index	
Poorest	3583 (17.9)
Poorer	4028 (20.1)
Middle	4135 (20.6)
Richer	4189 (20.9)
Richest	4094 (20.4)
No. of Household Members	
Less or equal 4	9661 (48.2)
4+	10,367 (51.8)
Mass Media exposure	
No	10,101 (50.4)
Yes	9928 (49.6)
Own a mobile phone	
No	6089 (30.4)
Yes	13,940 (69.6)
Currently residing with husband	
Living with her	15,999 (79.9)
Elsewhere living (within Bangladesh)	1361 (6.8)
Elsewhere living (not in Bangladesh)	1699 (8.5)
Missing	969 (4.8)
No. of under-5 children	
0	10,432 (52.1)
1	7468 (37.3)
1+	2129 (10.6)
No. of child (ever born)	
0	2094 (10.5)
1	4515 (22.5)
2	6467 (32.3)
3	4096 (20.4)
3+	2857 (14.3)
Area of residence	
Urban	5700 (28.5)
Rural	14,328 (71.5)
Division	
Barisal	1199 (6)
Chattogram	3749 (18.7)
Dhaka	5080 (25.4)
Khulna	2389 (11.9)
Mymensingh	1527 (7.6)
Rajshahi	2625 (13.1)
Rangpur	2291 (11.4)
Sylhet	1169 (5.8)
Number of decisions made by women	
0	3193 (15.9)
1	1817 (9.1)
2	2179 (10.9)
3	2823 (14.1)
4	10,017 (50)
Currently Pregnant	
No	18,848 (94.1)
Yes	1181 (5.9)
Currently breastfeeding	
No	15,955 (79.7)
Yes	4073 (20.3)

an additional 24% of them had symptoms of mild depression (Fig. 2). The distribution of study variables by GAD-7 and PHQ-9 severity categories are provided in Table S2 of the supplementary material B.

The prevalence of symptoms of moderate to severe anxiety and depression varied across different divisions. Regarding anxiety, the burden was significantly ($p < 0.05$) higher in Khulna (6%, 95% CI: 4.6%, 7.3%), Rangpur (6%, 95% CI: 4.6%, 7.2%), and Sylhet (6%, 95% CI: 4.3%, 7.5%) compared to the national average. Regarding depression, the burden was significantly ($p < 0.05$) higher in Rangpur (8%, 95% CI: 6.0%, 9.6%), Sylhet (7%, 95% CI: 4.7%, 8.8%), and Khulna (7%, 95% CI: 5.3%, 8.0%) than the national average (Fig. 3).

Table 2 presents the prevalence of moderate to severe anxiety and depressive symptoms across various demographic and socioeconomic factors, while Table 3 presents the AOR of the multivariable logistic regression model with 95% CI. Regarding respondent's age, it was found that the prevalence of moderate to severe anxiety and depression increased with age, women aged 45–49 exhibiting the highest prevalence of anxiety at 7% (95% CI: 5.5%, 7.8%) and depression at 7% (95% CI: 6.2%, 8.5%). In the model, older women was significantly associated with a higher likelihood of anxiety (AOR: 2.33, 95% CI: 1.55, 3.50) and depression (AOR: 1.63, 95% CI: 1.15, 2.31). Women with higher secondary education or above had the lowest prevalence of anxiety (3%, 95% CI: 2.0%, 3.4%) and depressive symptoms (3%, 95% CI: 2.5%, 4.0%). Higher education was associated with a lower likelihood of both anxiety (AOR: 0.60, 95% CI: 0.40, 0.90) and depressive symptoms (AOR: 0.67, 95% CI: 0.46, 0.96). Conversely, women with no formal education exhibited the highest prevalence of anxiety (6%, 95% CI: 5.2%, 7.2%) and depression (7%, 95% CI: 5.8%, 8.1%). In terms of occupation, women engaged in non-agricultural work had a higher prevalence of anxiety (6%, 95% CI: 4.5%, 7.2%) and were at greater likelihood of anxiety in the model (AOR: 1.40, 95% CI: 1.10, 1.78). In contrast, women involved in agricultural work had a lower prevalence of depressive symptoms (5%, 95% CI: 3.8%, 5.4%) and lower odds of depression (AOR: 0.74, 95% CI: 0.62, 0.88). Muslim women demonstrated a higher prevalence of moderate to severe anxiety (5%, 95%



Fig. 1 Item-wise symptomatology percentage of GAD-7 and PHQ-9 scale among reproductive-aged women in Bangladesh ($n = 20{,}029$)

CI: 4.1%, 5.1%) and depression (5%, 95% CI: 4.5%, 5.6%). In the model, women of other religions had a lower likelihood of anxiety (AOR: 0.50, 95% CI: 0.37, 0.69) and depressive symptoms (AOR: 0.66, 95% CI: 0.51, 0.86). Women with one under-five child had a relatively lower prevalence of anxiety (4%, 95% CI: 3.0%, 4.3%) and depressive symptoms (4%, 95% CI: 3.6%, 4.9%). In the model, these women had lower chances of both anxiety (AOR: 0.82, 95% CI: 0.68, 0.98) and depressive symptoms (AOR: 0.84, 95% CI: 0.71, 0.99). Lastly, women who made all four major household decisions had higher odds of anxiety (AOR: 1.30, 95% CI: 1.00, 1.69) and depressive symptoms (AOR: 1.41, 95% CI: 1.07, 1.80).

Among respondents who showed any symptoms of anxiety or depression, around 22% (95% CI: 19.9%, 25.0%) of respondents reported seeking care for anxiety or depression from healthcare providers. Only 8% (95% CI: 6.6%, 10.1%) of respondents reported taking medicine prescribed by a healthcare provider for anxiety or depression (Fig. 4).



Fig. 2 Severity of symptoms of anxiety and depression among women of reproductive age (15–49 years)

Discussion

Mental health, though crucial, is often overlooked and under-researched in Bangladesh. This study provides a national estimate of symptoms of anxiety and depression. The BDHS 2022 for the first time introduced a mental health module, revealing that seven in every hundred women experience moderate to severe anxiety or depression. Older women, those with lower education, and those making household decisions reported higher of these symptoms. Geographically, women in Khulna, Rangpur, and Sylhet experienced higher burden of these symptoms. Only 22% of women experiencing these symptoms sought help from a healthcare provider, and just 8% used medication.

Our findings align with those from Nepal and India, as well as the Global Burden of Disease (GBD) estimates. We found a prevalence of 4% for anxiety symptoms and 5% for depression. These figures closely match the recent GBD 2021 estimates, which indicate that globally, 7% (95% CI: 6 to 9) of reproductive-aged women have an anxiety disorder, and 4% (95% CI: 4 to 5) have a depressive disorder. In Bangladesh, the estimated prevalence is 6% (95% CI: 4 to 8) for anxiety and 7% (95% CI: 5 to 9) for depression (GBD 2021 Results 2022). While the GBD estimates are model-based, our community survey data may provide a more accurate reflection of the country's specific context. A study in Nepal found that 23% of women experienced anxiety and 6% had depressive symptoms. Using a cut-off of ≥ 10, Nepal's anxiety prevalence was 8%, which is still double our findings, potentially due to wealth-related disparities (Pandey et al. 2024). In India, anxiety disorders were most common among women aged 18–49, with a prevalence of 4%, followed by depressive disorders at 3% (Annajigowda et al. 2023). A systematic review of 24 studies during COVID-19 in Bangladesh reported a pooled prevalence of 47% for both depression and anxiety (Hosen et al. 2021). Another study indicated that 78% of women in urban slums experienced depressive symptoms and 59% anxiety symptoms within 18 months after a spontaneous abortion (Koly et al. 2023). The differences between our estimates and those from other Bangladeshi studies may be attributed to various factors, including the emergency context of COVID-19 and the high-risk populations facing significant stressors.

Mild symptoms of anxiety and depression, if left unmanaged, can escalate into more severe forms. Our study focused on moderate to severe cases, but we found that 22% of women reported mild anxiety symptoms, and 24% reported mild depressive symptoms. Previous research indicates that 14% of individuals with subthreshold anxiety progress to severe anxiety disorders within three years (Bosman et al. 2019). Similarly, a study on mental health patients found that the frequency of severe depressive episodes increased from 26% after the first episode to 50% by the fifteenth (Kessing 2008). It is crucial to recognize that mild symptoms can worsen over time without appropriate lifestyle modifications and counseling. While basic lifestyle changes and increased awareness often help manage mild symptoms, moderate to severe cases usually require more comprehensive interventions such as psychological counseling, therapy, or medication. In Bangladesh, awareness and support structures for

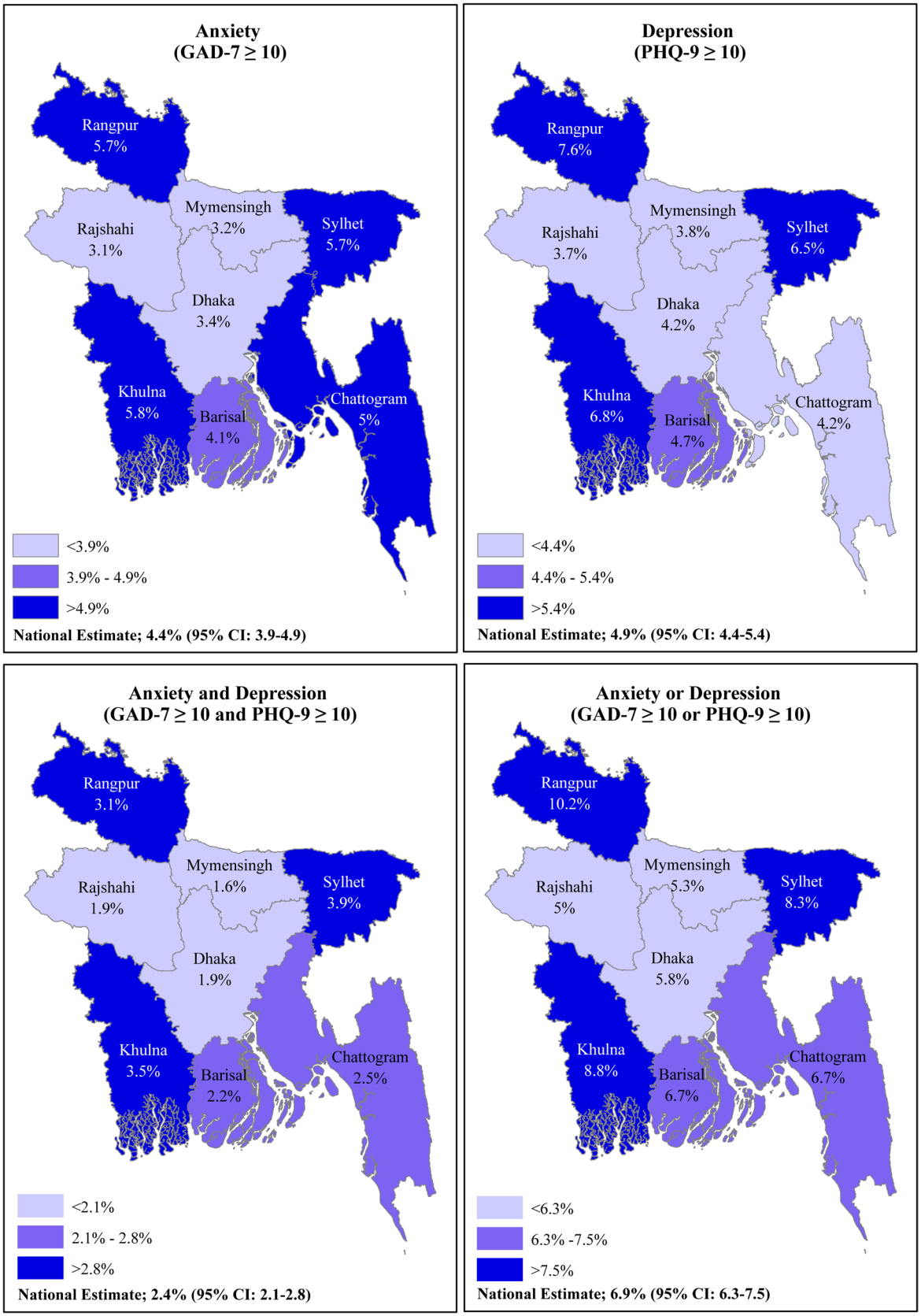


Fig. 3 Division specific prevalence of symptoms of anxiety, depression, both and either

Table 2 Distribution of prevalence of anxiety, depression, anxiety/depression and anxiety and depression co-occurrence across various demographic and socioeconomic factors among reproductive-aged women in Bangladesh ($n = 20,029$)

Variables	Anxiety symptoms (> = 10)	Depressive symptoms (> = 10)	Anxiety or depressive symptoms	Anxiety and depressive symptoms
	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)
Overall	**4.4 (3.9,4.9)**	**4.9 (4.4, 5.4)**	**6.9 (6.3,7.5)**	**2.4 (2.1,2.8)**
Women Age				
15–19	2.3 (1.6,3.5)	3.7 (2.7,4.9)	4.6 (3.5,6.1)	1.3 (0.8,2.3)
20–24	2.7 (2.1,3.5)	3.2 (2.6,4.0)	4.4 (3.7,5.4)	1.5 (1.1,2.0)
25–29	3.5 (2.9,4.3)	4.6 (3.9,5.5)	6 (5.2,7.0)	2.1 (1.6,2.7)
30–34	4.9 (4.1,5.8)	5.2 (4.3,6.2)	7.3 (6.3,8.4)	2.8 (2.2,3.6)
35–39	5.3 (4.5,6.3)	5.4 (4.6,6.4)	7.9 (6.9,9.1)	2.8 (2.2,3.5)
40–44	5.4 (4.4,6.5)	5.4 (4.4,6.5)	8.1 (6.8,9.5)	2.7 (2.0,3.5)
45–49	6.5 (5.5,7.8)	7.3 (6.2,8.5)	9.9 (8.6,11.4)	3.9 (3.1,4.9)
Women Education				
No education	6.1 (5.2,7.2)	6.9 (5.8,8.1)	9.6 (8.4,11.0)	3.3 (2.6,4.2)
Primary	4.9 (4.2,5.6)	5.1 (4.4,5.9)	7.3 (6.5,8.2)	2.7 (2.2,3.3)
Secondary	4.1 (3.6,4.7)	4.8 (4.2,5.4)	6.4 (5.7,7.2)	2.4 (2.0,3.0)
Higher secondary or above	2.6 (2.0,3.4)	3.2 (2.5,4.0)	4.8 (3.9,5.8)	1 (0.7,1.5)
Husbands' Age				
15–19	2.2 (0.5,8.4)	4.4 (1.6,11.3)	5.3 (2.2,12.3)	1.2 (0.2,8.3)
20–24	1.8 (1.1,3.1)	3.5 (2.4,4.9)	4.5 (3.3,6.2)	0.8 (0.4,1.7)
25–29	3.2 (2.4,4.3)	3.8 (2.9,4.9)	5.1 (4.1,6.3)	1.9 (1.3,2.9)
30–34	2.9 (2.2,3.7)	3.9 (3.0,5.0)	5 (4.1,6.2)	1.7 (1.2,2.4)
35–39	3.6 (2.9,4.4)	4.5 (3.7,5.5)	6 (5.1,7.1)	2.1 (1.5,2.8)
40–44	4.3 (3.5,5.3)	4.3 (3.6,5.2)	6.3 (5.3,7.4)	2.3 (1.8,3.0)
45–49	4.4 (3.6,5.4)	4.9 (4.0,6.1)	6.7 (5.6,8.0)	2.6 (2.0,3.5)
50 +	6 (5.2,7.0)	6.2 (5.4,7.2)	9.1 (8.1,10.2)	3.1 (2.6,3.8)
Missing				
Husbands' Education				
No education	5.3 (4.5,6.2)	6.3 (5.4,7.3)	8.6 (7.6,9.8)	2.9 (2.3,3.7)
Primary	4.2 (3.5,5.0)	4.7 (4.0,5.6)	6.4 (5.5,7.4)	2.5 (2.0,3.2)
Secondary	3.8 (3.2,4.6)	4.4 (3.8,5.2)	6.2 (5.5,7.1)	2 (1.6,2.5)
Higher secondary or above	2.7 (2.1,3.5)	3 (2.4,3.7)	4.3 (3.5,5.1)	1.4 (1.0,2.0)
Missing				
Women's Occupation				
Housewife	4.2 (3.7,4.7)	4.9 (4.3,5.5)	6.7 (6.0,7.4)	2.4 (2.0,2.8)
Agricultural	4.4 (3.6,5.2)	4.5 (3.8,5.4)	6.6 (5.7,7.6)	2.3 (1.8,3.0)
Non-agricultural	5.7 (4.5,7.2)	5.6 (4.5,7.0)	8.3 (6.9,10.0)	3 (2.1,4.2)
Officer/Businessmen	4.4 (3.1,6.3)	5 (3.3,7.6)	7.1 (5.1,9.8)	2.3 (1.4,3.9)
Retired/Others	5.3 (2.4,11.1)	9.8 (5.4,16.9)	11.7 (7.0,18.9)	3.3 (1.1,9.5)
Husbands' Occupation				
Not working	8.7 (6.5,11.5)	8.7 (6.5,11.6)	11.4 (8.9,14.4)	6 (4.2,8.6)
Agricultural	3.9 (3.3,4.7)	4.8 (4.1,5.7)	6.6 (5.7,7.5)	2.2 (1.7,2.8)
Non-agricultural	4 (3.4,4.6)	4.6 (4.0,5.3)	6.5 (5.8,7.4)	2 (1.6,2.6)
Officer/Businessmen	3.6 (3.0,4.3)	4 (3.5,4.7)	5.5 (4.8,6.3)	2.2 (1.7,2.7)
Retired/Others	8.3 (3.4,19.0)	6.1 (2.5,13.9)	10.3 (4.8,20.7)	4.1 (1.3,12.4)
Missing				
Marital status				
Currently married	4.1 (3.6,4.6)	4.7 (4.2,5.2)	6.5 (5.9,7.1)	2.3 (1.9,2.7)
Separated/Deserted/ Divorced/Widowed	10.6 (8.7,12.9)	10.2 (8.4,12.4)	15 (12.7,17.5)	5.8 (4.5,7.6)

Table 2 (continued)

Variables	Anxiety symptoms (> = 10)	Depressive symptoms (> = 10)	Anxiety or depressive symptoms	Anxiety and depressive symptoms
	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)
Religion				
Islam	4.5 (4.1,5.1)	5 (4.5,5.6)	7.1 (6.5,7.7)	2.5 (2.1,2.9)
Others	2.7 (2.0,3.8)	3.8 (2.8,5.0)	4.8 (3.7,6.1)	1.7 (1.1,2.6)
Wealth Index				
Poorest	5.1 (4.2,6.1)	5.6 (4.7,6.7)	7.9 (6.8,9.1)	2.9 (2.3,3.6)
Poorer	4.1 (3.3,5.1)	5.1 (4.2,6.2)	7 (6.0,8.1)	2.3 (1.6,3.1)
Middle	4.6 (3.7,5.6)	5.4 (4.4,6.5)	7.1 (6.0,8.3)	2.9 (2.2,3.8)
Richer	4.6 (3.8,5.5)	4.6 (3.8,5.5)	6.6 (5.7,7.7)	2.5 (1.9,3.2)
Richest	3.6 (2.9,4.4)	4 (3.3,4.8)	5.9 (5.0,7.0)	1.7 (1.2,2.3)
No. of Household Members				
Less or equal 4	4.7 (4.2,5.3)	5.4 (4.8,6.0)	7.4 (6.7,8.2)	2.7 (2.3,3.2)
4+	4.1 (3.5,4.7)	4.5 (4.0,5.1)	6.4 (5.7,7.1)	2.2 (1.8,2.7)
Mass Media exposure				
No	4.4 (3.8,5.1)	4.9 (4.3,5.6)	6.9 (6.1,7.7)	2.5 (2.0,3.0)
Yes	4.3 (3.8,4.9)	5 (4.4,5.6)	6.9 (6.2,7.6)	2.4 (2.0,2.9)
Own a mobile phone				
No	4.1 (3.5,4.7)	5.2 (4.5,5.9)	6.9 (6.1,7.7)	2.4 (2.0,2.8)
Yes	4.5 (4.0,5.1)	4.8 (4.3,5.4)	6.9 (6.2,7.6)	2.5 (2.1,2.9)
Currently residing with husband				
Living with her	4.1 (3.7,4.6)	4.8 (4.3,5.3)	6.6 (6.0,7.2)	2.2 (1.9,2.6)
Elsewhere living (within Bangladesh)	4 (2.9,5.5)	4.7 (3.6,6.1)	6 (4.6,7.7)	2.7 (1.9,3.8)
Elsewhere living (not in Bangladesh)	3.7 (2.6,5.4)	3.7 (2.6,5.2)	5.4 (3.9,7.4)	2 (1.2,3.2)
Missing				
No. of under-5 children				
0	5.1 (4.5,5.7)	5.7 (5.1,6.3)	8 (7.4,8.8)	2.7 (2.3,3.2)
1	3.6 (3.0,4.3)	4.2 (3.6,4.9)	5.7 (5.0,6.5)	2.1 (1.7,2.7)
1 +	3.7 (2.8,4.9)	3.6 (2.8,4.8)	5.2 (4.1,6.6)	2.1 (1.5,3.1)
No. of child (ever born)				
0	3.5 (2.7,4.5)	4.7 (3.7,6.0)	6.2 (5.0,7.6)	2.1 (1.5,2.9)
1	3.3 (2.7,3.9)	4.1 (3.5,4.9)	5.4 (4.7,6.2)	2 (1.6,2.5)
2	4 (3.5,4.7)	4.5 (3.9,5.2)	6.3 (5.6,7.1)	2.2 (1.8,2.6)
3	4.9 (4.1,5.7)	5.6 (4.8,6.6)	7.8 (6.9,8.9)	2.7 (2.1,3.4)
3 +	6.8 (5.7,8.2)	6.4 (5.4,7.5)	9.5 (8.2,11.1)	3.7 (2.9,4.6)
Area of residence				
Urban	4 (3.3,4.9)	4.4 (3.6,5.3)	6.4 (5.5,7.5)	2 (1.5,2.7)
Rural	4.5 (4.0,5.1)	5.1 (4.6,5.8)	7 (6.4,7.8)	2.6 (2.2,3.1)
Division				
Barisal	4.1 (3.1,5.4)	4.7 (3.7,6.1)	6.7 (5.4,8.3)	2.2 (1.6,2.9)
Chittagong	5 (3.6,6.8)	4.2 (3.0,5.9)	6.7 (5.1,8.7)	2.5 (1.5,4.0)
Dhaka	3.4 (2.7,4.4)	4.2 (3.2,5.3)	5.8 (4.7,7.1)	1.9 (1.3,2.7)
Khulna	5.8 (4.6,7.3)	6.5 (5.3,8.0)	8.8 (7.4,10.4)	3.5 (2.6,4.8)
Mymensingh	3.2 (2.3,4.3)	3.8 (2.9,4.9)	5.3 (4.2,6.8)	1.6 (1.1,2.5)
Rajshahi	3.1 (2.3,4.1)	3.7 (3.0,4.7)	5 (4.1,6.1)	1.9 (1.2,2.7)
Rangpur	5.7 (4.6,7.2)	7.6 (6.0,9.6)	10.2 (8.5,12.3)	3.1 (2.2,4.4)
Sylhet	5.7 (4.3,7.5)	6.5 (4.7,8.8)	8.3 (6.3,10.9)	3.9 (2.8,5.4)

Table 2 (continued)

Variables	Anxiety symptoms (> = 10)	Depressive symptoms (> = 10)	Anxiety or depressive symptoms	Anxiety and depressive symptoms
	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)
Number of decisions made by women				
0	5.4 (4.6,6.4)	6 (5.1,7.1)	8.6 (7.4,9.9)	2.9 (2.3,3.6)
1	3.8 (2.7,5.1)	4.8 (3.8,6.2)	6.4 (5.1,7.9)	2.2 (1.5,3.3)
2	3.3 (2.5,4.3)	4 (3.1,5.1)	5.5 (4.5,6.8)	1.7 (1.2,2.5)
3	4.3 (3.5,5.3)	5 (4.1,6.0)	6.9 (6.0,8.1)	2.4 (1.8,3.1)
4	4.4 (3.8,5.1)	4.8 (4.1,5.5)	6.7 (5.9,7.5)	2.5 (2.0,3.1)
Currently Pregnant				
No	4.4 (4.0,4.9)	4.9 (4.5,5.5)	6.9 (6.3,7.4)	2.5 (2.1,2.9)
Yes	4.2 (3.0,5.9)	4.9 (3.6,6.5)	7.1 (5.5,9.1)	2 (1.3,3.2)
Currently breastfeeding				
No	4.6 (4.2,5.2)	5.3 (4.7,5.8)	7.4 (6.8,8.0)	2.5 (2.2,3.0)
Yes	3.3 (2.7,4.1)	3.6 (3.0,4.5)	4.9 (4.1,5.9)	2 (1.5,2.6)

mental health issues are limited. Women with mild symptoms are unlikely to be identified by the health system or receive specialized care due to the lack of mental health services (Brenda J. Butler 2008). Consequently, these women may remain undiagnosed and unattended, potentially leading to a progression from mild to more severe mental health conditions if timely intervention is not provided.

Women in the Khulna, Rangpur, and Sylhet divisions are more likely to experience symptoms of anxiety and depression compared to those in Dhaka. These regions are more vulnerable to natural disasters, such as flooding and droughts, which can cause significant stress and trauma, contributing to a higher burden of mental health issues (Stanke et al. 2012). A study in the disaster-prone Koyra Upazila of Khulna found that 63% of women reported moderate to severe depression (Moyna et al. 2024). This situation reflects a complex interplay of environmental, socio-economic, and healthcare access factors that contribute to mental health disparities in these areas. Women in Dhaka tend to have better access to education and employment opportunities, while the lack of such opportunities in Khulna, Rangpur, and Sylhet can lead to feelings of hopelessness and despair, further increasing the risk of anxiety and depression (Boreham and Schutte 2023).

Older women, particularly those aged 45–49, are more likely to experience symptoms of anxiety and depression than younger women aged 15–19. This finding aligns with a recent systematic review indicating that older women face a higher risk of mental illness (Kiely et al. 2019). Contributing factors may include biological causes, comorbidities, and the effects of menopause (Siegel and Mathews 2015). Comorbidities such as arthritis, diabetes, or heart disease, can further exacerbate mental health symptoms, making older women more vulnerable to depression and anxiety (Daré et al. 2019). These chronic conditions often lead to a reduced quality of life, and increased stress, which can contribute to the higher burden of depression and anxiety among older women. These conditions also frequently require ongoing medical care and management, which can add to the emotional burden and stress, further intensifying mental health symptoms. Menopause, in particular, is associated with a significant risk of mood disorders due to declining estrogen levels, which imbalance neurotransmitters like serotonin (Harun et al. 2020; Alblooshi et al. 2023). Also, a recent study in Bangladesh highlighted that limited knowledge about menopause, compounded by cultural stigma, shame, and lack of familial or social support, contributes to depressive symptoms and poor coping among older women (Harun et al. 2020). Studies show that the pandemic further increased anxiety and depression among older women (Vahia et al. 2020).

We found that women with higher education levels, such as secondary education or above, have a lower likelihood of experiencing anxiety and depression. Previous studies support this, showing a positive impact of education on mental well-being (Courtin et al. 2019; Jiang et al. 2020). For instance, a study among European women revealed that those with higher education levels had significantly lower odds of anxiety and depressive symptoms (Bjelland et al. 2008; Chlapecka et al. 2023). Education provides access to resources, improved coping mechanisms, and a stronger sense of control, which can reduce anxiety (Reshi et al. 2022).

Our results show having at least one child aged under five seemed to reduce the odds of both anxiety and depression among reproductive-aged women. Similar to our findings, the Harvard Study of Moods and Cycles revealed women with at least one child had a 30% decreased odds of past

Table 3 Associated factors of depression, anxiety, and anxiety/depression among reproductive-aged women in Bangladesh ($n = 20,029$)

Variables	Anxiety symptoms		Depressive symptoms		Anxiety or depressive symptoms		Anxiety and depressive symptoms	
	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value
Women Age								
15–19	**Ref**		**Ref**		**Ref**		**Ref**	
20–24	1.13 (0.76, 1.67)	0.543	0.92 (0.66, 1.28)	0.612	0.97 (0.73, 1.3)	0.846	1.09 (0.65, 1.82)	0.746
25–29	1.47 (1, 2.14)	**0.048**	1.33 (0.97, 1.82)	0.077	1.34 (1.02, 1.78)	**0.037**	1.51 (0.92, 2.47)	0.102
30–34	1.97 (1.36, 2.85)	**< 0.001**	1.42 (1.04, 1.95)	**0.027**	1.53 (1.16, 2.02)	**0.002**	2.02 (1.25, 3.27)	**0.004**
35–39	2.08 (1.43, 3.02)	**< 0.001**	1.33 (0.96, 1.83)	0.082	1.54 (1.16, 2.03)	**0.003**	1.94 (1.18, 3.17)	**0.009**
40–44	2.08 (1.4, 3.09)	**< 0.001**	1.28 (0.91, 1.81)	0.157	1.53 (1.13, 2.06)	**0.005**	1.82 (1.08, 3.09)	**0.026**
45–49	2.33 (1.55, 3.5)	**< 0.001**	1.63 (1.15, 2.31)	**0.006**	1.71 (1.25, 2.33)	**0.001**	2.62 (1.55, 4.45)	**< 0.001**
Women Education								
No education	**Ref**		**Ref**		**Ref**		**Ref**	
Primary	0.89 (0.7, 1.12)	0.321	0.86 (0.69, 1.07)	0.173	0.83 (0.69, 1.01)	0.058	0.98 (0.72, 1.34)	0.892
Secondary	0.86 (0.67, 1.11)	0.25	0.88 (0.7, 1.12)	0.304	0.83 (0.67, 1.01)	0.065	1 (0.72, 1.41)	0.982
Higher secondary or above	0.6 (0.4, 0.9)	**0.012**	0.67 (0.46, 0.96)	**0.032**	0.72 (0.53, 0.98)	**0.034**	0.4 (0.22, 0.72)	**0.002**
Husbands' Education								
No education	**Ref**		**Ref**		**Ref**		**Ref**	
Primary	0.94 (0.76, 1.15)	0.534	0.83 (0.69, 1.01)	0.064	0.84 (0.71, 1)	**0.046**	0.98 (0.75, 1.28)	0.862
Secondary	0.92 (0.73, 1.15)	0.454	0.81 (0.66, 1)	**0.047**	0.86 (0.72, 1.03)	0.104	0.84 (0.62, 1.13)	0.251
Higher secondary or above	0.78 (0.56, 1.08)	0.14	0.61 (0.45, 0.83)	**0.002**	0.62 (0.48, 0.81)	**< 0.001**	0.89 (0.58, 1.37)	0.596
Women's Occupation								
Housewife	**Ref**		**Ref**		**Ref**		**Ref**	
Agricultural	0.89 (0.74, 1.08)	0.233	0.74 (0.62, 0.88)	**0.001**	0.83 (0.71, 0.97)	**0.016**	0.75 (0.58, 0.96)	**0.023**
Non-agricultural	1.4 (1.1, 1.78)	**0.006**	1 (0.78, 1.28)	0.984	1.15 (0.94, 1.41)	0.171	1.25 (0.9, 1.74)	0.176
Officer/Businessmen	1.35 (0.89, 2.05)	0.156	1.36 (0.92, 2)	0.121	1.34 (0.96, 1.87)	0.087	1.44 (0.83, 2.5)	0.193
Retired/Others	1.22 (0.56, 2.66)	0.613	2.04 (1.15, 3.6)	**0.014**	1.73 (1.01, 2.94)	**0.044**	1.56 (0.61, 3.99)	0.349
Husbands' Occupation								
Housewife	**Ref**		**Ref**		**Ref**		**Ref**	
Agricultural	0.41 (0.3, 0.57)	**< 0.001**	0.49 (0.36, 0.67)	**< 0.001**	0.52 (0.39, 0.69)	**< 0.001**	0.32 (0.21, 0.47)	**< 0.001**
Non-agricultural	0.48 (0.36, 0.66)	**< 0.001**	0.56 (0.42, 0.76)	**< 0.001**	0.62 (0.47, 0.81)	**< 0.001**	0.35 (0.24, 0.51)	**< 0.001**
Officer/Businessmen	0.48 (0.35, 0.67)	**< 0.001**	0.54 (0.39, 0.74)	**< 0.001**	0.55 (0.41, 0.73)	**< 0.001**	0.43 (0.29, 0.64)	**< 0.001**
Retired/Others	0.92 (0.42, 1.99)	0.829	0.64 (0.26, 1.54)	0.316	0.86 (0.42, 1.74)	0.671	0.62 (0.22, 1.79)	0.379
Religion								
Islam	**Ref**		**Ref**		**Ref**		**Ref**	
Others	0.5 (0.37, 0.69)	**< 0.001**	0.66 (0.51, 0.86)	**0.002**	0.58 (0.46, 0.74)	**< 0.001**	0.59 (0.4, 0.87)	**0.008**
Wealth Index								
Poorest	**Ref**		**Ref**		**Ref**		**Ref**	
Poorer	0.85 (0.67, 1.07)	0.166	1.03 (0.83, 1.28)	0.789	0.98 (0.81, 1.18)	0.809	0.85 (0.62, 1.16)	0.296
Middle	0.96 (0.75, 1.22)	0.717	1.14 (0.91, 1.42)	0.26	1.03 (0.85, 1.25)	0.739	1.09 (0.8, 1.49)	0.576
Richer	1 (0.78, 1.28)	0.991	0.98 (0.77, 1.25)	0.868	0.98 (0.8, 1.21)	0.854	1.01 (0.73, 1.4)	0.956
Richest	0.79 (0.59, 1.07)	0.131	0.9 (0.68, 1.2)	0.474	0.94 (0.74, 1.2)	0.634	0.62 (0.41, 0.94)	**0.024**
No. of Household Members								
Less or equal 4	**Ref**		**Ref**		**Ref**		**Ref**	
4+	0.93 (0.8, 1.09)	0.364	0.9 (0.78, 1.04)	0.153	0.95 (0.84, 1.07)	0.394	0.82 (0.67, 1.01)	0.067
No. of under-5 children								
0	**Ref**		**Ref**		**Ref**		**Ref**	
1	0.82 (0.68, 0.98)	**0.03**	0.84 (0.71, 1)	**0.046**	0.78 (0.67, 0.9)	**0.001**	0.99 (0.78, 1.26)	0.963
1 +	0.96 (0.71, 1.28)	0.776	0.77 (0.58, 1.03)	0.082	0.78 (0.61, 0.99)	0.045	1.11 (0.75, 1.65)	0.585
Number of decisions made by women								
0	**Ref**		**Ref**		**Ref**		**Ref**	

Table 3 (continued)

Variables	Anxiety symptoms		Depressive symptoms		Anxiety or depressive symptoms		Anxiety and depressive symptoms	
	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value
1	1.18 (0.84, 1.67)	0.332	1.17 (0.87, 1.58)	0.303	1.1 (0.85, 1.43)	0.471	1.45 (0.91, 2.29)	0.117
2	1.06 (0.76, 1.5)	0.72	0.98 (0.72, 1.33)	0.899	0.97 (0.75, 1.26)	0.832	1.17 (0.73, 1.87)	0.519
3	1.26 (0.92, 1.71)	0.147	1.14 (0.87, 1.5)	0.356	1.12 (0.88, 1.42)	0.347	1.43 (0.94, 2.18)	0.094
4	1.3 (1, 1.69)	**0.042**	1.41 (1.07, 1.8)	**0.04**	1.09 (0.89, 1.33)	0.424	1.57 (1.09, 2.26)	**0.016**
Currently breastfeeding								
No	**Ref**		**Ref**		**Ref**		**Ref**	
Yes	1.11 (0.88, 1.39)	0.395	0.96 (0.78, 1.19)	0.729	0.99 (0.82, 1.19)	0.901	1.13 (0.84, 1.53)	0.408
Area of residence								
Urban	**Ref**		**Ref**		**Ref**		**Ref**	
Rural	1.08 (0.89, 1.3)	0.425	1.11 (0.92, 1.32)	0.273	1.06 (0.91, 1.23)	0.46	1.2 (0.93, 1.54)	0.173
Division								
Dhaka	**Ref**		**Ref**		**Ref**		**Ref**	
Barishal	1.23 (0.86, 1.75)	0.253	1.13 (0.81, 1.57)	0.464	1.21 (0.91, 1.59)	0.19	1.11 (0.68, 1.81)	0.677
Chittagong	1.54 (1.22, 1.94)	**< 0.001**	1.03 (0.82, 1.3)	0.79	1.2 (0.99, 1.46)	0.06	1.42 (1.04, 1.96)	**0.03**
Khulna	1.79 (1.39, 2.31)	**< 0.001**	1.69 (1.34, 2.14)	**< 0.001**	1.62 (1.32, 1.99)	**< 0.001**	2.14 (1.54, 2.97)	**< 0.001**
Mymensingh	0.88 (0.61, 1.26)	0.475	0.87 (0.62, 1.21)	0.396	0.91 (0.69, 1.2)	0.496	0.77 (0.46, 1.27)	0.309
Rajshahi	0.83 (0.62, 1.13)	0.233	0.87 (0.66, 1.14)	0.322	0.82 (0.64, 1.03)	0.092	0.97 (0.66, 1.44)	0.888
Rangpur	1.74 (1.34, 2.27)	**< 0.001**	2 (1.58, 2.53)	**< 0.001**	2.01 (1.64, 2.46)	**< 0.001**	1.61 (1.12, 2.31)	**0.01**
Sylhet	1.43 (1.02, 2.01)	**0.036**	1.47 (1.08, 1.99)	**0.014**	1.35 (1.03, 1.77)	**0.03**	1.77 (1.16, 2.7)	**0.008**

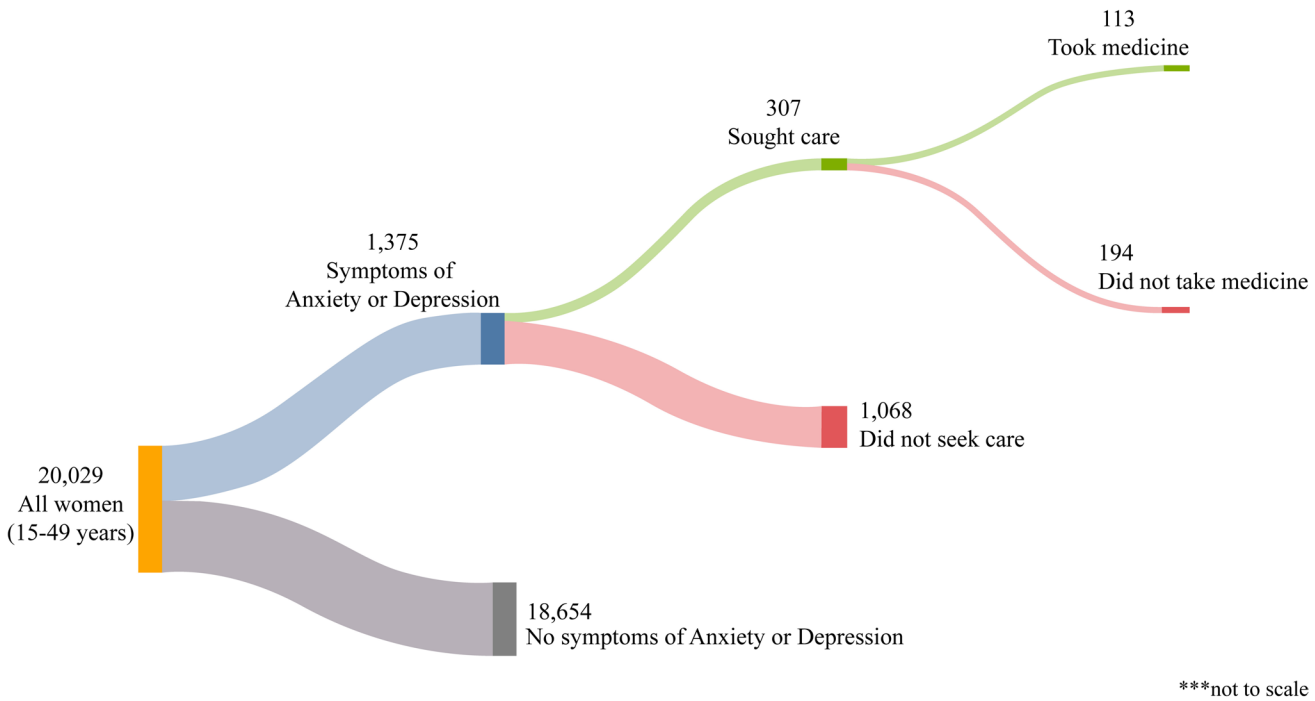


Fig. 4 Care-seeking practice among women with symptoms of anxiety and depression in Bangladesh ($n = 1375$)

mood disorder compared to nulliparous women and the risk reduction was greater among women with three or more children (Harlow et al. 1999). Another cross-sectional study conducted among 376 women attending clinics in Pakistan reported that having at least 2 children was protective against depression compared to having more than two (Ali and Zuberi 2012). Caring for a young child often creates a sense of purpose and a clear routine, which can foster a feeling of control and stability in daily life. Additionally, the close emotional bond that develops between a mother and her child can provide social and emotional support, helping to buffer against feelings of isolation, a common contributor to mental health issues. Furthermore, interactions with the child, family, and community may create more opportunities for social connection, which is known to reduce the likelihood of anxiety and depression. Having more than one child was associated with elevated scores on several postpartum stressors, including overload and working mother concerns (Walker and Murry 2022). However, parous women did not show significant differences in depression, and anxiety levels compared to nulliparous women (Bhavsar and Thakar 2020). Women with multiple children developed various coping strategies to manage postpartum mood disorders, including taking time alone or with others, managing emotions and thoughts, and maintaining a manageable workload (Robertson et al. 2013; Walker and Murry 2022). Positive coping strategies and higher social support were associated with lower perceived stress levels (Goletzke et al. 2017).

Women involved in making all four major household decisions were more likely to experience symptoms of anxiety and depression. This finding aligns with a study in Uzbekistan, which found that women with decision-making autonomy had higher depressive symptoms (Hadley et al. 2010; Vahia et al. 2020). Conversely, a community survey in India indicated that lower autonomy was associated with a higher likelihood of common mental disorders (CMD) (Patel et al. 2005; Siegel and Mathews 2015). Further exploration is needed to clarify the relationship between mental health and women's empowerment in the context of Bangladesh. Our results suggest that women's involvement in family decision-making might challenge traditional gender norms, potentially leading to increased stress and worsening mental health. Alternatively, empowered women may be more willing to share their mental health challenges, indicating a possible reverse causation effect.

About 22% of women experiencing symptoms of anxiety or depression sought care, with only 8% using medication. This is consistent with a previous study in Nepal, where 18% of women sought help (Shawon et al. 2024), reflecting low care-seeking rates for mental health in other low- and middle-income countries (Rathod et al. 2017). Mental healthcare in Bangladesh is severely inadequate, with no structured services at primary or secondary levels (Hossain et al. 2014). The country has only 260 psychiatrists, 565 psychologists, and 700 nurses serving a population of 162 million, limiting access to mental health services (Harlow et al. 1999; WHO 2019). The National Institute of Mental Health facilitates 200 beds, while another psychiatric hospital in Pabna offers 500 beds, of which only 31 are allocated for in-patients, representing a mere 8% of total hospital beds (Hasan et al. 2021). Overall, about 92% of adults with mental disorders do not receive treatment (Hasan et al. 2021). Cultural stigma and low mental health literacy further hinder care-seeking behavior (Kessing 2008; Ali and Zuberi 2012). Additionally, the maldistribution of resources in urban areas, a lack of effective mental health policies, and biases against women and marginalized groups prevent many from accessing treatment (Kiely et al. 2019; Koly et al. 2023). To improve this situation, investing in preventative mental health initiatives at a societal level is essential which includes integrating mental health services into primary healthcare, developing community-based mental health programs, and implementing school-based mental health education to build resilience early on. Additionally, promoting self-help measures and strengthening digital mental health solutions, such as mobile applications for screening and counseling, and expanding telepsychiatry services can be helpful.

Strengths and limitations

Our study is the first to present the burden of mental disorders in Bangladesh using a nationally representative sample of 20,029 women from the BDHS. It employed a rigorous sampling framework and robust data collection methods, ensuring high-quality data through thorough monitoring and quality checks. The survey used the validated Bangla versions of the GAD-7 and PHQ-9 scales to assess anxiety and depression symptoms, enhancing data accuracy. In contrast to the 2019 National Mental Health Survey (NMHS), which involved 3,805 women aged 18–99, our study's larger sample size allows for more precise and representative estimates. The DHS design also collects comprehensive demographic and health data, whereas the smaller NMHS sample limits the validity and generalizability of its findings.

However, there are limitations. As this is the first BDHS with a mental health module, we can only provide current estimates without trend data. The cross-sectional design restricts our ability to establish causality or explore temporal relationships; a longitudinal study would offer greater insight. Additionally, relying on self-reported data may introduce biases, such as social desirability and recall bias, which are common in surveys using these tools. While our study measured key factors associated with mental health symptoms, some relevant variables could not be included. For instance, experiences of traumatic events such as

natural disasters, menopausal status, and attitudes toward help-seeking were not measured, limiting a comprehensive understanding of all potential contributors to mental health outcomes. Regarding care-seeking practices, while we report whether respondents sought help and whether this was from healthcare providers, we were unable to capture the quality of care or specific types of care sought (e.g., counseling or psychiatric services). Additionally, data on non-pharmacological treatments such as therapy were not collected, meaning our findings may not fully reflect the range of treatment modalities utilized by respondents. Lastly, information on medication use was limited to recent pharmacological treatments (past 2 weeks) and did not capture long-term use, further restricting the scope of our analysis.

Conclusion

Mental disorders, especially anxiety and depression, emerge as a public health challenge in Bangladesh, with heightened prevalence observed among older women, those engaged in household decision-making, and residents of Rangpur, Khulna, and Sylhet. The issue is further compounded by a low rate of mental health care-seeking behaviors. There is an urgent need for the development and implementation of targeted interventions and policies to address regional disparities and support high-risk populations. Enhancing public awareness and integrating mental health services into primary healthcare systems are critical strategies for improving care-seeking behavior and treatment adherence, thereby advancing mental well-being among women in Bangladesh.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00737-025-01564-3.

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Data Availability The datasets used in this study are owned by the Demographic and Health Surveys (DHS) Program. These data are available upon request and approval from https://dhsprogram.com/data/.

Declarations

Ethics approval and consent to participate We obtained permission to access this dataset and conduct this research from the Demographic and Health Survey's Program following standard protocol. As we conducted a secondary analysis, no additional ethical approval was required to conduct this study. The protocols and questionnaires of DHS surveys have been reviewed and approved by the ICF Institutional Review Board (IRB) and the IRB of the host countries. Ethical review documentation is available upon request from the Protecting the Privacy of DHS Survey Respondents.

Competing interest The authors have no competing interests to declare that are relevant to the content of this article.

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References

Acharya DR, Bell JS, Simkhada P et al (2010) Women's autonomy in household decision-making: A demographic study in Nepal. Reprod Health 7. https://doi.org/10.1186/1742-4755-7-15

Afifi M (2007) Gender differences in mental health. Singapore Med J 48:385–391. https://smj.sma.org.sg/4805/4805ra1.pdf

Akter E, Hossain AT, Rahman AE et al (2023) Levels and determinants of quality antenatal care in Bangladesh: Evidence from the Bangladesh Demographic and Health Survey. PLoS One 18 https://doi.org/10.1371/journal.pone.0269767

Alblooshi S, Taylor M, Gill N (2023) Does menopause elevate the risk for developing depression and anxiety? Results from a systematic review. Australas Psychiatr 31. https://doi.org/10.1177/10398562231165439

Ali FA, Zuberi RW (2012) Association of sociodemographic factors with depression in women of reproductive age. Asia Pac J Public Health 24. https://doi.org/10.1177/1010539510364668

Annajigowda HH, Nirisha LP, Ganjekar S, et al (2023) Common mental disorders among women in reproductive age group: An analysis of national mental health survey, India 2016. Indian J Psychiatry 65. https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry_832_23

Banik R, Islam MdS, Ahmed M et al (2022) General psychiatric symptoms among Bangladeshi people approximately one year after the onset of the COVID-19 pandemic. BMC Psychiatry 22:615. https://doi.org/10.1186/s12888-022-04232-3

Bhavsar M, Thakar J (2020) Stress, depression, anxiety and coping strategies in parous and nulliparous women. Int J Indian Psychol 8:264–268

Bjelland I, Krokstad S, Mykletun A et al (2008) Does a higher educational level protect against anxiety and depression? The HUNT study. Soc Sci Med 66. https://doi.org/10.1016/j.socscimed.2007.12.019

Boreham ID, Schutte NS (2023) The relationship between purpose in life and depression and anxiety: A meta-analysis. J Clin Psychol 79. https://doi.org/10.1002/jclp.23576

Bosman RC, ten Have M, de Graaf R et al (2019) Prevalence and course of subthreshold anxiety disorder in the general population: A three-year follow-up study. J Affect Disord 247. https://doi.org/10.1016/j.jad.2019.01.018

Butler BJ (2008) Psychiatric disorders in women. In: Clouse Amy L, Sherif Katherine (eds) Current clinical practice. Humana Press, Totowa

Chlapecka A, Wolfová K, Fryčová B, Cermakova P (2023) Educational attainment and anxiety in middle-aged and older Europeans. Sci Rep 13. https://doi.org/10.1038/s41598-023-40196-4

Courtin E, Nafilyan V, Glymour M et al (2019) Long-Term effects of compulsory schooling on physical, mental and cognitive ageing: A natural experiment. J Epidemiol Community Health (1978) 73. https://doi.org/10.1136/jech-2018-211746

Croft TN, Marshall AMJ, Allen CK (2020) Guide to DHS statistics - DHS-7 (version 2). USAID. Retrieved from https://www.dhsprogram.com/pubs/pdf/DHSG1/Guide_to_DHS_Statistics_DHS-7_v2.pdf

Daré LO, Bruand PE, Gérard D et al (2019) Co-morbidities of mental disorders and chronic physical diseases in developing and emerging countries: A meta-analysis. BMC Public Health 19. https://doi.org/10.1186/s12889-019-6623-6

Dutta GK, Sarker BK, Ahmed HU et al (2022) Mental healthcare-seeking behavior during the perinatal period among women in rural Bangladesh. BMC Health Serv Res 22. https://doi.org/10.1186/s12913-022-07678-z

Ferdausi F, Al-Zubayer MA, Keramat SA, Ahammed B (2022) Prevalence and associated factors of underweight and overweight/obesity among reproductive-aged women: A pooled analysis of data from South Asian countries (Bangladesh, Maldives, Nepal and Pakistan). Diabetes Metab Syndr: Clin Res Rev 16. https://doi.org/10.1016/j.dsx.2022.102428

GBD 2019 Mental Disorders Collaborators (2022) Global, regional, and national burden of 12 mental disorders in 204 countries and territories, 1990–2019: a systematic analysis for the global burden of disease study 2019. Lancet Psychiatry 9:137–150. https://doi.org/10.1016/S2215-0366(21)00395-3

GBD 2021 Forecasting Collaborators (2024) Burden of disease scenarios for 204 countries and territories, 2022–2050: a forecasting analysis for the global burden of disease study 2021. Lancet 403:2204–2256. https://doi.org/10.1016/s0140-6736(24)00685-8

GBD 2021 Results (2022) Tool: use the following to cite data included in this download: global burden of disease collaborative network. Global burden of disease study 2021 (GBD 2021) Results. Institute for Health Metrics and Evaluation (IHME), Seattle

Goletzke J, Kocalevent RD, Hansen G et al (2017) Prenatal stress perception and coping strategies: Insights from a longitudinal prospective pregnancy cohort. J Psychosom Res 102. https://doi.org/10.1016/j.jpsychores.2017.09.002

Goyal S, Gupta B, Sharma E et al (2020) Psychiatric morbidity, cultural factors, and health-seeking behaviour in perinatal women: A cross-sectional study from a tertiary care centre of North India. Indian J Psychol Med 42. https://doi.org/10.4103/IJPSYM.IJPSYM_96_19

Hadley C, Brewis A, Pike I (2010) Does less autonomy erode women's health? Yes. no. maybe. Am J Hum Biol 22. https://doi.org/10.1002/ajhb.20959

Harlow BL, Cohen LS, Otto MW et al (1999) Prevalence and predictors of depressive symptoms in older premenopausal women: The Harvard Study of Moods and Cycles. Arch Gen Psychiatry 56. https://doi.org/10.1001/archpsyc.56.5.418

Harun MGD, Salema U, Chowdhury A et al (2020) Knowledge and attitudes associated with menopause among women aged 45 to 60 years: A pilot study among rural and urban women in Bangladesh. Menopause 27. https://doi.org/10.1097/GME.0000000000001525

Hasan MT, Anwar T, Christopher E et al (2021) The current state of mental healthcare in Bangladesh: part 2 – setting priorities. BJPsych Int 18. https://doi.org/10.1192/bji.2021.42

Hosen I, al-Mamun F, Mamun MA (2021) Prevalence and risk factors of the symptoms of depression, anxiety, and stress during the COVID-19 pandemic in Bangladesh: a systematic review and meta-analysis. Glob Ment Health 8. https://doi.org/10.1017/gmh.2021.49

Hossain MD, Ahmed HU, Chowdhury WA et al (2014) Mental disorders in Bangladesh: A systematic review. BMC Psychiatry 14. https://doi.org/10.1186/s12888-014-0216-9

Islam A (2015) Mental health and the health system in Bangladesh: situation analysis of a neglected domain. Am J Psychiatr Neurosci 3. https://doi.org/10.11648/j.ajpn.20150304.11

Jiang W, Lu Y, Xie H (2020) Education and mental health: Evidence and mechanisms. J Econ Behav Organ 180. https://doi.org/10.1016/j.jebo.2020.09.032

Jorm AF (2000) Mental health literacy: public knowledge and beliefs about mental disorders. Br J Psychiatr 177:396–401

Kessing LV (2008) Severity of depressive episodes during the course of depressive disorder. Br J Psychiatr 192. https://doi.org/10.1192/bjp.bp.107.038935

Kiely KM, Brady B, Byles J (2019) Gender, mental health and ageing. Maturitas 129:76–84. https://doi.org/10.1016/j.maturitas.2019.09.004

Koly KN, Saba J, Billah MA et al (2023) Depressive symptoms and anxiety among women with a history of abortion living in urban slums of Bangladesh. BMC Psychol 11. https://doi.org/10.1186/s40359-023-01224-0

Kroenke K, Spitzer RL, Williams JBW (2001) The PHQ-9: Validity of a brief depression severity measure. J Gen Intern Med 16:606–613. https://doi.org/10.1046/j.1525-1497.2001.016009606.x

Moyna SA, Hasan K, Kabir KH et al (2024) Depressive symptoms among women in disaster-prone region in Bangladesh. J Affect Disord Rep 16:100762. https://doi.org/10.1016/j.jadr.2024.100762

NIPORT and ICF (2024) Bangladesh demographic and health survey 2022: final report. Dhaka, Bangladesh, and Rockville, Maryland. https://www.dhsprogram.com/pubs/pdf/FR386/FR386.pdf

Pandey AR, Adhikari B, Bista B et al (2024) Prevalence, determinants and care-seeking behaviour for anxiety and depression in Nepalese population: a secondary analysis of data from Nepal Demographic and Health Survey 2022. BMJ Open 14:e078582. https://doi.org/10.1136/bmjopen-2023-078582

Patel V, Kirkwood B, Weiss H et al (2005) Chronic fatigue in developing countries: Population based survey of women in India. Br Med J 330:1190

Patel V, Saxena S, Lund C et al (2018) The Lancet Commission on global mental health and sustainable development. The Lancet 392:1553–1598

Rathod S, Pinninti N, Irfan M et al (2017) Mental health service provision in low-and middle-income countries. Health Serv Insights 10:1178632917694350

Reshi IA, Sudha T, Dar SA (2022) Women's access to education and its impact on their empowerment: a comprehensive review. MORFAI J 1. https://doi.org/10.54443/morfai.v1i2.760

Robertson P, Curtis R, Lasher R et al (2013) Experiences of postpartum mood disorders of women with more than one child. Fam J 21. https://doi.org/10.1177/1066480713491068

Rutstein SO (2015) Steps to constructing the new DHS wealth index. ICF International, Rockville 6

Shawon MSR, Hossain FB, Ahmed R et al (2024) Role of women empowerment on mental health problems and care-seeking

behavior among married women in Nepal: secondary analysis of nationally representative data. Arch Womens Ment Health 27. https://doi.org/10.1007/s00737-024-01433-5

Siegel AM, Mathews SB (2015) Diagnosis and treatment of anxiety in the aging woman. Curr Psychiatry Rep 17:93. https://doi.org/10.1007/s11920-015-0636-3

Spitzer RL, Kroenke K, Williams JBW, Lo B (2006) A brief measure for assessing generalized anxiety disorder. Arch Intern Med 166:1092–1097. https://doi.org/10.1001/archinte.166.10.1092

Stanke C, Murray V, Amlôt R et al (2012) The effects of flooding on mental health: Outcomes and recommendations from a review of the literature. PLoS Curr. https://doi.org/10.1371/4f9f1fa9c3cae

The Lancet Global Health (2020) Mental health matters. Lancet Glob. Health 8:e1352. https://doi.org/10.1016/S2214-109X(20)30432-0

USAID (2023) The Demographic and health surveys program. In: 2025. https://www.usaid.gov/global-health/demographic-and-health-surveys-program. Accessed 13 Jan 2025

Vahia IV, Jeste DV, Reynolds CF III (2020) Older adults and the mental health effects of COVID-19. JAMA 324:2253–2254. https://doi.org/10.1001/jama.2020.21753

von Elm E, Altman DG, Egger M et al (2014) The Strengthening the reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. Int J Surg 12:1495–1499. https://doi.org/10.1016/j.ijsu.2014.07.013

Wahid SS, Raza WA, Mahmud I, Kohrt BA (2023) Climate-related shocks and other stressors associated with depression and anxiety in Bangladesh: a nationally representative panel study. Lancet Planet Health 7. https://doi.org/10.1016/S2542-5196(22)00315-1

Walker LO, Murry N (2022) maternal stressors and coping strategies during the extended postpartum period: a retrospective analysis with contemporary implications. Women's Health Reports 3. https://doi.org/10.1089/whr.2021.0134

Whiteford HA, Degenhardt L, Rehm J et al (2013) Global burden of disease attributable to mental and substance use disorders: findings from the global burden of disease study 2010. The Lancet 382. https://doi.org/10.1016/S0140-6736(13)61611-6

World Health Organization (2019a) Bangladesh WHO Special initiative for mental health situational assessment. https://www.who.int/publications/m/item/bangladesh---who-special-initiative-for-mental-health. Accessed 9 Jan 2025

World Health Organization (2019b) Mental health. https://www.who.int/health-topics/mental-health#tab=tab_2. Accessed 19 May 2024

World Health Organization (2022a) World mental health report: transforming mental health for all. https://www.who.int/publications/i/item/9789240049338

World Health Organization (2022b) Mental disorders. https://www.who.int/news-room/fact-sheets/detail/mental-disorders. Accessed 6 Sep 2024

ZabunNahar Md, Sohan KF, Supti, et al (2022) Prevalence and associated risk factors for mental health problems among female university students during COVID-19 pandemic: A cross-sectional study findings from Dhaka, Banglades. Heliyon 8:e10890

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